

Nifty100 Investment Analytics - Analyst Guide

Author: Karun Reddy

Version: 1.0

Project Overview

End-to-end investment analytics platform for Nifty100 companies using Python, SQLite, Pandas, FastAPI and Streamlit.

Architecture

ETL loads Excel data into SQLite. Analytics compute ratios, rankings, screening, valuation, cash-flow metrics and reports. FastAPI exposes REST endpoints. Streamlit provides an interactive dashboard.

Key Database Tables

companies, financial_ratios, scorecard, rankings, valuation_summary, risk_analysis, investment_scores, investor_reports, company_summary, stock_prices.

Analytics

ROE, ROCE, Net Profit Margin, Debt-to-Equity, Interest Coverage, CAGR, Cash Flow KPIs, investment scoring, peer comparison and valuation.

Dashboard

8-screen Streamlit dashboard with screening, portfolio, valuation, sector, company and report views.

FastAPI

REST endpoints for companies, documents, screener, peers, sectors, portfolio, valuation, risk, investment scores, investor reports, company summary, dashboard, stock prices and analytics.

Testing

Pytest suite executed successfully with 77 passing tests. HTML report generated.

Run Commands

```
streamlit run src/dashboard/app.py  
uvicorn src.api.main:app --reload
```

Conclusion

The project provides an end-to-end investment analytics workflow from ETL to API and dashboard.